Rotaract Club of Xiamen University Malaysia

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Nexgen
Project / Solution Name	NexFlow (Orbyte Smart Yield) — Domain 2
Name (Leader)	Yeo Jin Hong
Email & Phone (Leader)	kadenyeojinhong@gmail.com / 0143987675
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	https://nexflow-kadenyeojinhong.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Nexgen-XMUM/Nexflow

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect and (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

NexFlow is a local hackathon prototype. A vanilla HTML/CSS/JavaScript browser workspace supports staff, owner and customer demo personas. The Python FastAPI/Pydantic backend validates closing stock counts, calculates clearance prices and food-footprint estimates, drafts WhatsApp-style promotions using an offline template or optional DeepSeek API, and exposes surprise-box quotation, simulated payment, pickup and donation handover endpoints. SQLite stores temporary demo batches, orders, quotes, loyalty points and donation statuses; a JSON catalogue supplies products and prices. The demo launcher provides a simulated store clock and resets its temporary database on restart. No payment, WhatsApp message, or charity contact is actually sent. Separately, the team fine-tuned an Ultralytics YOLOv8 bread detector (best.pt); a standalone Python app and ONNX Runtime Web experiment can run detection. The final welcome page uses precomputed detection boxes as a front-end animation, not live inference or stock synchronization. Code is shared through GitHub; its main FastAPI demonstration runs locally, not as a public production service.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.13; HTML / CSS / JavaScript	Main backend and browser UI	Python: PSF; web standards
FastAPI 0.156.4; Starlette 1.9.0; Uvicorn 0.54.0	REST endpoints / ASGI demo server	MIT; BSD-3-Clause
Pydantic 2.14.1; HTTPX 0.28.1; python-dotenv 1.2.1	Data validation, optional AI API, config	MIT; BSD-3-Clause
sqlite3 (stdlib); pytest 9.1.1	Local data ledger; automated testing	PSF; MIT
Ultralytics 8.4.183; PyTorch; OpenCV; NumPy	Separate YOLOv8 model training/inference	AGPL-3.0; BSD-3; Apache-2.0
ONNX Runtime Web 1.17.3; Tailwind CSS; React / Express / Vite (exploratory Replit UI)	Browser YOLO experiment; UI and separate draft	MIT (main packages); verify extras

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek Chat Completions API (optional)	AI-assisted promotional copy when configured; offline template otherwise	Team/developer key; tier to confirm
WhatsApp Business API — not used	WhatsApp conversation is a local animation, not messaging	N/A
Payment gateway — not used	Checkout is a local simulated payment	N/A
Roboflow Hosted Inference API — not used	Bread dataset was sourced; local YOLO prototype uses model files	N/A
None / N/A	—	—

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
GitHub (Microsoft)	Public source repository and version control	Public repository; plan to confirm
Replit (development only)	Leader-supplied experimental React/Express UI project; not the FastAPI deployment	Workspace tier/region to confirm
Local machine / SQLite (not cloud)	Final FastAPI demo and temporary local database	Localhost, Malaysia time UTC+08
Google Fonts / public UI CDNs	Front-end fonts and optional Tailwind / ONNX Runtime JavaScript assets	Public web resources; free tier
None / N/A	—	—

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Bread detector: YOLOv8 best.pt (15 classes)	Ultralytics base + team fine-tuning	Fine-tuned; separate Python YOLO prototype	AGPL-3.0 (Ultralytics); confirm distribution compliance
Bread detector: ONNX export	Exported from team fine-tuned YOLO weights	Fine-tuned model; browser inference experiment	Ultralytics / AGPL-3.0 upstream
DeepSeek chat model (optional)	DeepSeek API	External pretrained model; promo copy only if enabled	DeepSeek API Terms; not default demo
None / N/A	—	—	—
None / N/A	—	—	—

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Bread Detector	https://universe.roboflow.com/breaddetector/bread-detector	Public object-detection dataset	CC BY 4.0	N
Sample bakery catalogue / demo batch profiles	Team-authored JSON and fixed demo fixtures	Synthetic / team-generated	Team-created	N
Five bread demonstration images	From the Bread Detector project; impeccable/demo_images	Public sample images	CC BY 4.0	N
None / N/A	—	—	—	—
None / N/A	—	—	—	—

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✓ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
ChatGPT (OpenAI; model/version varied)	Planning, architecture review, code review, content and documentation	Major
DeepSeek Work / coding agent (version unrecorded)	Frontend integration, i18n, tests and bug fixes	Major
OpenAI Codex (version unrecorded)	Code assistance, backend development and Git workflows	Major
Claude / Claude Artifacts (version unrecorded)	Browser YOLO prototype and UI design experiments	Moderate
Replit development/AI tools (leader to verify)	Separate alternative React/Express interface prototype	Extent to confirm

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-assisted contributions were inspected against the intended workflows and API contracts, and checked for unintended changes before merging. The team ran Python pytest and Node frontend tests plus browser walkthroughs covering welcome, stock counting, promotional approval, simulated box purchasing, pickup, food rescue and the results dashboard. The team distinguishes simulated behavior from working integrations and retains human approval over code merges and public claims. These checks are suitable for a hackathon prototype, not a security or production certification.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✓ Yes No (provisional — team leader must confirm)

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Runpod	Already redeemed voucher from Runpod, but isa is different, if run in Runpod after run out of credit I can't training and fix in our computer.	Runpod voucher is claimed
Replit — sponsorship status unverified	Experimental separate UI workspace, not required to run submitted FastAPI code	Leader to confirm any sponsored access/credits
None / N/A	—	—
None / N/A	—	—

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No sponsor-provided infrastructure is required for the core NexFlow demonstration: its backend, temporary SQLite data and promo-template fallback run locally. Replit was involved in a separate prototype, but whether any official SDC sponsor credits or resources were used has not been verified. If the team received or used official sponsor resources, change the answer above to Yes and list the sponsor, resource and purpose before the leader signs.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The competition demo uses fictional staff/customer personas and synthetic store data. It has no production registration or real payment processing. A fixed demo user ID, quotes, simulated orders and donation events are kept in a temporary local SQLite database and cleared when the demo server restarts. Browser localStorage may hold language/theme settings and temporary purchase state until cleared. Demo login inputs are not sent to a real authentication service or stored as passwords. If optional DeepSeek promotional generation is enabled, product and promotion details may be sent to that provider; users should not include personal customer data. The application has no production-grade authentication, encryption-at-rest or formal data retention policy, and should not process actual customer information.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The homepage shows fixed, precomputed YOLOv8 detection boxes, not live inference. The separate trained model may misclassify bread or return low-confidence results; staff review is necessary. Optional DeepSeek promotions may hallucinate or fail because of availability or account limits, so a deterministic offline template is available and the owner reviews copy before simulated sending. Safety, storage and allergen facts in the blind-box workflow are synthetic demo values, not verified food-safety checks. Payments, WhatsApp broadcasts and charity handovers are simulations; carbon figures are estimates, not certified emissions reductions. There is no real account authentication. If a customer triggers a quote before staff register leftovers, auto-seeded stock can cause a 409 conflict; the recommended demo sequence registers leftovers before quoting. This prototype is not ready for public production use.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✓ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✓ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✓ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✓ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✓ Any use of official SDC sponsor resources has been declared in Section 5.
- ✓ The prototype described here is accessible to judges through the links provided in Section 1.
- ✓ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature (Team Leader): _____
Date: _____